FDA Adverse Events Milestone II Project Proposal

Introduction and overview

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We will develop a reproducible pharmacovigilance pipeline that (1) predicts whether an incoming FDA Adverse Event Reporting System (FAERS) report indicates hospitalization or death and (2) discovers recurring drug-reaction phenotypes without using outcome labels. Rather than treating the supervised and unsupervised analyses as isolated modeling exercises, we will integrate them into a single pipeline that accepts newly acquired FAERS data, applies standardized preprocessing and feature engineering, generates calibrated serious-outcome risk predictions, assigns reports to recurring safety-pattern clusters, and produces structured outputs that can be reviewed or visualized. The project will support report triage and hypothesis generation, not causal attribution.

As a consumable solution, the pipeline could supplement existing pharmacovigilance workflows used by pharmaceutical companies or regulatory organizations such as the FDA. The supervised component could help prioritize newly received reports for further review based on their predicted probability of hospitalization or death, while the unsupervised component could group reports with similar drug-reaction patterns to identify recurring or emerging safety phenotypes for further investigation. Combining these components creates a reproducible workflow in which the same preprocessing, prediction, clustering, and reporting procedures can be applied to future FAERS data rather than existing only as analyses presented in the final report. The final project will document the pipeline architecture, required inputs, processing steps, model outputs, and potential integration into pharmacovigilance review processes.

FAERS is a spontaneous-reporting system with underreporting, missing data, stimulated reporting, duplicates and follow-up reports, uncertain drug-event attribution, and no exposure denominator. Therefore, the pipeline cannot estimate adverse-event incidence or establish that a drug caused an event. Its outputs should instead be interpreted as decision-support signals that help prioritize reports and generate hypotheses requiring additional clinical or epidemiological evaluation.

Zhao et al. (2023), “Identifying the serious clinical outcomes of adverse reactions to drugs by a multi-task deep learning framework,” used FAERS and a graph-based multi-task model to classify seven serious outcomes. Our project differs by using a time-based train/test split, interpretable logistic-regression and gradient-boosting models, probability calibration, explicit leakage checks, and report-level clustering to reveal safety-pattern archetypes. This is new work and does not reuse supervised or unsupervised learning from a prior credited project.

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Part A (Supervised Learning)

FDA FAERS (FDA Adverse Event Reporting Systems)/ AEMS (Adverse Event Monitoring System) public reports from 2020-2025, accessed through the openFDA Drug Adverse Event Quarterly Data Extract. Available fields include report dates and sources, patient demographics, drugs and roles, indications, routes, reactions, and seriousness outcomes.

We will compare multiple supervised model types, including logistic regression and random forest. Logistic regression will provide an interpretable baseline that allows us to examine how individual features relate to predicted serious outcomes, while gradient boosting will allow us to capture nonlinear relationships and interactions that may not be represented well by a linear model. For our feature representations, we will test several approaches appropriate for the mixture of structured and text-based FAERS data. Numeric variables will be scaled to place measurements on comparable ranges, categorical variables will be one-hot encoded so they can be incorporated without imposing artificial numerical relationships, and missingness indicators will be included because patterns of missing data may themselves contain useful information. Sparse term frequency–inverse document frequency (TF-IDF) representations of reaction and indication terms will allow the models to capture clinically relevant patterns while reducing the influence of very common terms. Outcome fields, post-outcome information, and direct outcome-proxy terms will be excluded to prevent data leakage and ensure predictions are based only on information that would reasonably be available when a report is evaluated.

RxNorm and openFDA drug annotations will normalize brand names to active ingredients; FAERS already uses MedDRA preferred terms for reactions. Preparation requires retaining the latest case version, creating one report-level record, standardizing ages and weights, encoding multi-valued fields, and consolidating rare categories.

Outcome: We will report PR-AUC as the primary metric plus ROC-AUC, precision, recall, F1, confusion matrices, Brier score, and calibration. Visuals will include outcome prevalence over time, precision-recall and calibration curves, confusion matrices, and coefficient.

Part B (Unsupervised Learning)

We will use the same cleaned 2020-2025 openFDA FAERS adverse-event reports, with seriousness labels withheld during model fitting.

What recurring drug-reaction patterns are identified across the dataset? Which drugs, adverse reactions, and indications characterize each identified group? How do the prevalence of these groups and their eventual serious-outcome composition change over time?

We will need to deduplicate case versions, normalize ingredients, form one report-level observation, construct drug-reaction-indication matrices, remove ultra-rare terms and outcome proxies, and scale or weight features appropriately.

We will compare truncated singular value decomposition (SVD) followed by k-means clustering with hierarchical clustering based on cosine distance, using either term frequency-inverse document frequency (TF-IDF) features or binary report-level features. Because hierarchical clustering can become computationally impractical for very large datasets, it will be performed on a representative subsample, while k-means will be used to cluster the full dataset. Uniform Manifold Approximation and Projection (UMAP) will be used only to create two-dimensional visualizations and not as primary evidence that the identified clusters are valid.

RxNorm and openFDA annotations will support ingredient normalization; MedDRA terms supplied in FAERS will provide standardized reaction concepts.

Outcome: We will evaluate cluster quality by measuring how cohesive and well-separated the clusters are using the silhouette score. We will also examine whether cluster sizes are reasonably balanced, whether similar clusters appear across different random seeds and bootstrap samples, and whether each cluster can be meaningfully interpreted from its most characteristic terms. Seriousness outcomes will be compared across clusters only after the clusters have been created, ensuring that outcome information does not influence the clustering process.

A UMAP scatterplot, cluster-by-reaction heatmap, top-term bar charts, and temporal cluster-prevalence plot will communicate separation, composition, and change.

Team Planning

Orko Mookerjee: domain framing, data cleaning and deduplication, supervised modeling, leakage audit, and final narrative.

Che-Yu Liu: literature review, external mappings, unsupervised modeling, and stability analysis.
Nikhil Jain: evaluation, visualization, reproducibility review, and presentation.

All members: code review, interpretation, and final editing.

Project Timeline

Project Component	Primary Role / Responsibility	Timeline	Expected Deliverable
Project scope and literature review	Orko Mookerjee, Nikhil Jain, Che-Yu Liu	Week 1	Finalized research questions, project scope, and supporting literature

Data acquisition and schema	Orko Mookerjee	Week 1	FAERS data acquired through the application programming interface (API) and organized into the planned data schema
Exploratory data analysis (EDA)	Orko Mookerjee, Nikhil Jain, Che-Yu Liu	Weeks 1–2	Initial assessment of distributions, missingness, class balance, and data-quality concerns
Data preprocessing and quality auditing	Orko Mookerjee	Weeks 1–3	Normalized, deduplicated, and quality-checked analysis dataset
Supervised learning	Nikhil Jain, Che-Yu Liu	Weeks 2–4	Trained models for predicting hospitalization or death, including logistic regression and random forest
Unsupervised learning	Orko Mookerjee, Che-Yu Liu	Weeks 2–4	Report-level clustering of recurring drug-reaction patterns and assessment of cluster characteristics
Model evaluation and validation	Orko Mookerjee, Nikhil Jain, Che-Yu Liu	Weeks 4–5	Temporal validation, error analysis, probability calibration, and clustering stability assessment
Visualization and interpretation	Orko Mookerjee, Nikhil Jain, Che-Yu Liu	Weeks 4–5	Visualizations of model performance, cluster characteristics, and changes over time
Report and presentation development	Che-Yu Liu	Weeks 4–6	Drafted and revised written report and presentation

Final review and delivery	Nikhil Jain	Week 6	Feedback incorporated and final report, presentation, and reproducible pipeline completed